Jet Diffuser with Nozzles JB



Description
Aluminum linear diffuser with individually steerable nozzles. Recommended for use in traffic areas, cinemas, museums, theaters, etc.

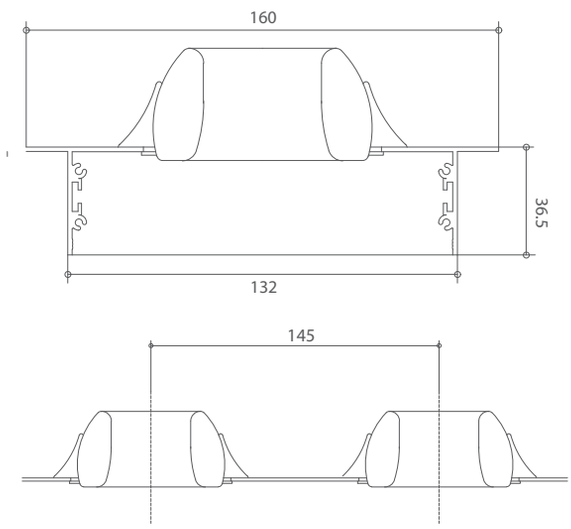
Application
Various air supply systems

Characteristics
The diffuser is produced, depending on technical and architectural requirements, with one or two rows of nozzles.
Nozzles have a diameter of 40 mm for air passage and are available in two colors: white RAL 9016 and black RAL 9005.
Due to the spherical design of the nozzle, the air flow can be directed into desired areas.
Allows unlimited airflow direction settings.

Accessories
Attachment spring **A**.
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Polypropylene nozzles

Technical design



Number of nozzles	Standard height
2	1row of nozzles = 132 mm 2 rows of nozzles = 294 mm
3	
4	
5	
6	
7	
8	
9	
10	

Assembly Methods
Attachment spring.

Selection table JB

Air flow m³/h	Number of nozzles	3	5	6	8	10
	Ak(m²)	0.0038	0.0063	0.0076	0.01	0.0126
120	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	9.5 5.7 2.8 8.77 < 20 43.0				
150	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	11.8 7.1 3.5 10.96 22.0 68.0				
200	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	15.8 9.5 4.7 14.62 29.0 121.0	12.2 7.3 3.7 8.82 < 20 43.0	11.2 6.7 3.3 7.31 < 20 30.0		
250	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	19.7 11.8 5.9 18.27 35.0 189.0	15.3 9.2 4.6 11.02 24.0 68.0	13.9 8.4 4.2 9.14 < 20 47.0	12.1 7.2 3.6 6.94 < 20 27.0	
300	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	23.7 14.2 7.1 21.93 40.0 272.0	18.3 11.0 5.5 13.23 29.0 98.0	16.7 10.0 5.0 10.96 25.0 68.0	14.5 8.7 4.3 8.33 < 20 38.0	
350	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	27.6 16.6 8.3 25.58 44.0 370.0	21.4 12.8 6.4 15.43 33.0 133.0	19.5 11.7 5.9 12.79 29.0 92.0	16.9 10.1 5.1 9.72 22.0 52.0	15.1 9.1 4.5 7.72 < 20 33.0
400	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		24.4 14.7 7.3 17.64 36.0 174.0	22.3 13.4 6.7 14.62 32.0 121.0	19.3 11.6 5.8 11.11 26.0 68.0	17.3 10.4 5.2 8.82 21.0 43.0
500	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		> 30 18.3 9.2 22.05 42.0 272.0	27.9 16.7 8.4 18.27 38.0 189.0	24.1 14.5 7.2 13.89 32.0 106.0	21.6 13.0 6.5 11.02 27.0 68.0
600	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		> 30 22.0 11.0 26.46 47.0 391.0	> 30 20.1 10.0 21.93 43.0 272.0	29.0 17.4 8.7 16.67 37.0 153.0	25.9 15.6 7.8 13.23 32.0 98.0
700	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			> 30 23.4 11.7 25.58 47.0 370.0	> 30 20.3 10.1 19.44 41.0 208.0	> 30 18.1 9.1 15.43 36.0 133.0
800	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				> 30 23.2 11.6 22.22 44.0 272.0	> 30 20.7 10.4 17.64 39.0 174.0
1000	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					> 30 25.9 13.0 22.05 45.0 272.0

Correction table for more rows

Number of rows	1	2
FNR	0	3
Fx	1	1.4

The values "X" will be corrected by multiplying with the coefficients from the table and "NR" through addition.

Annotation

*Ak [m²] - Free surface of the diffuser
*Veff[m/s] - Air velocity in diffuser
*NR [dB(A)] - Noise level without the room attenuation
*ΔPt [pa] - Pressure loss

X0.3 - Air jet length at speed Vk = 0.3 m/s
X0.5 - Air jet length at speed Vk = 0.5 m/s
X1.0 - Air jet length at speed Vk = 1.0 m/s

Order codes

Type - Number of rows	+	ADP IZ/N	/	Number of nozzles	/	Color of nozzles	/	O
JB 1 2		Insulated/Non-insulated adapter		On request		White RAL 9016 Black RAL 9005		**0** Painted white RAL 9016 **R** Other RAL paints